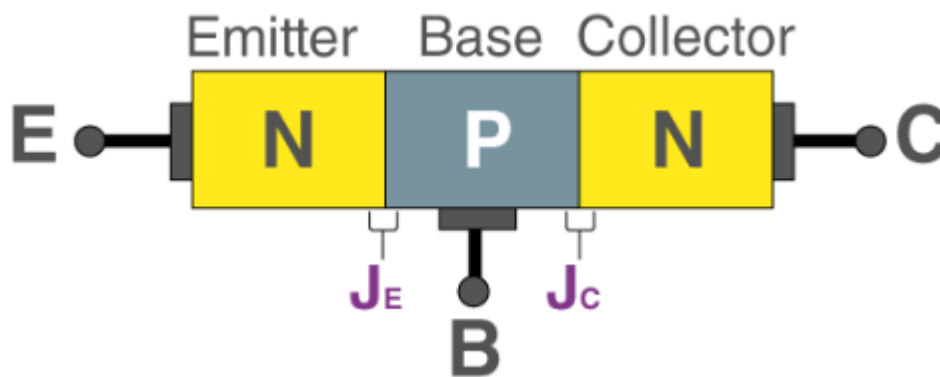


22GE004- BASICS OF ELECTRONICS ENGINEERING

UNIT 3 – BJT WORKING PRINCIPLES

1. INTRODUCTION

A Bipolar Junction Transistor (BJT) is a three-terminal device which consists of two PN-junctions formed by sandwiching either p-type or n-type semiconductor material between a pair of opposite type semiconductors. The primary function of BJT is to increase the strength of a weak signal, i.e., it acts as an amplifier. A BJT can also be used as a solid state switch in electronic circuits.



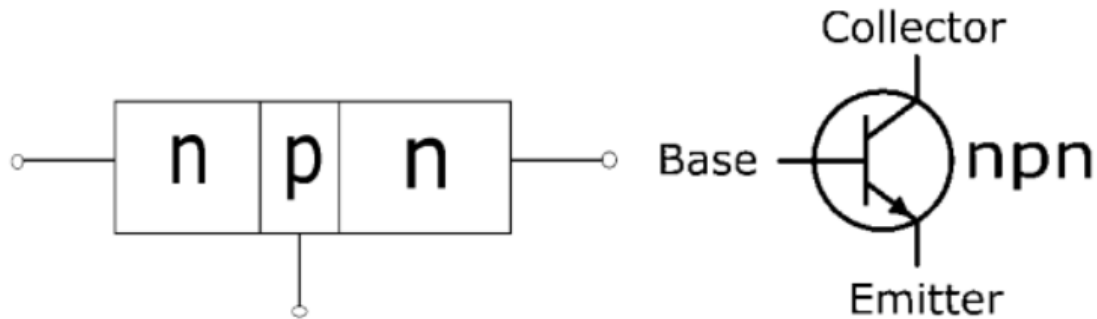
1.1. Types of BJT:

- NPN Transistor
- PNP Transistor

1.1.1 NPN Transistor

An NPN-transistor is composed of two n-type semiconductor materials which are separated by a thin layer of p-type semiconductor. The two terminals viz. Emitter and Collector are taken out from the two n-type semiconductor and the Base terminal is from the p-type semiconductor.

In BJT symbol, the arrow on the emitter terminal indicates the direction of conventional current in the emitter with forward bias. For NPN-transistor, the conventional current flows out of the emitter as indicated by the outing arrow.



1.1.2. PNP Transistor

A PNP-transistor is composed of two p-type semiconductors which are separated by a thin layer of n-type material. The two terminals viz. Emitter and Collector are taken out from the two p-type semiconductor layers and the Base terminal is from the n-type semiconductor. For the PNP-transistor, the conventional current flows into the emitter as indicated by the inward arrow.

1.2 IMPORTANT FEATURES OF BJT

- ✓ There are two PN-junctions, hence a transistor may be regarded as a combination of two back-to-back connected diodes.
- ✓ The collector region is wider than both emitter and base. The base is much thinner than both emitter and collector. During the transistor operation, a lots of heat is generated at the collector, hence the collector is made larger to dissipate the heat.
- ✓ A transistor has three sections of doped semiconductors. The one section is called the Emitter, the other is called the Collector, and the middle section is called the Base and forms two PN-junctions between emitter and collector.
- ✓ In general, the emitter-base junction of the BJT is made forward-biased, whereas the collector-base junction is reverse-biased.
- ✓ The resistance of forward-biased junction is very small as compared to that of the reverse-biased junction.
- ✓ The emitter is heavily doped so that it can supply a greater number of charge carriers (electrons or holes) to the base. The base is lightly doped and very thin, hence it passes most of the charge carriers injected by the emitter to the collector. The doping concentration of the collector region is moderate.

1.2. OPERATION OF BIPOLAR JUNCTION TRANSISTOR

There are three operating regions of a bipolar junction transistor:

- Active region: The region in which the transistors operate as an amplifier.
- Saturation region: The region in which the transistor is fully on and operates as a switch such that collector current is equal to the saturation current.
- Cut-off region: The region in which the transistor is fully off and collector current is equal to zero.

1.2 WORKING PRINCIPLE OF BJT

- ✓ Bipolar junction transistor is classified into two types namely PNP and NPN based on types of doping of the terminals. In NPN transistor, two semiconductor junctions are there which have a thin anode region doped with 'P' whereas PNP transistor includes two semiconductor junctions that include a thin cathode region doped with 'N'.
- ✓ The flow of charge within a transistor is because of the charge carrier's diffusion among the two sections which belongs to different concentrations of charge carriers. The emitter section is doped highly as compared to the remaining layers.
- ✓ Both the layers like base and collector include the same concentrations of charge carriers. So, between these three junctions, the BE junction can be connected in forwarding bias & the BC junction is reverse biased. The operating of BJT can be done in three different regions like active, saturation, and cut-off.

Active Region

- ✓ In this region, one junction is connected in a forward bias whereas the other one is connected in reverse bias. Here, the base current (I_b) is used to control the quantity of I_c (collector current). So, the active region can be mainly used for the purpose of amplification wherever this transistor performs like an amplifier through a gain ' β ' with the following equation;

$$I_c = \beta \times I_b$$

- ✓ The active region is also called a linear region which lies in between the two regions like the cutoff & the saturation. The typical operation of this transistor happens within this region.

Saturation Region

- ✓ In this region, the transistors both the junctions are connected in forwarding bias. So this region is mainly used for the ON-state for a switch wherever;
- ✓ $I_c = I_{sat}$
- ✓ Here, 'Isat' is the saturation current and it is the highest quantity of current flow among the two terminals like emitter as well as collector once this transistor is connected in the saturation region. As these junctions are connected in forwarding bias then the transistor works like a short circuit.

Cutoff Region

- ✓ In this region, both the transistor junctions are connected in reverse bias. Here this transistor works like the off condition of a switch wherever
 $I_c = 0$
- ✓ In the cut-off region, the operation is totally reverse as compared to the saturation region. So, there are no exterior supplies connected. If there is no collector current, then there is no emitter current.
- ✓ In this method, the transistor works like an off-state of the switch, and this mode can be achieved by decreasing base voltage to below the voltage of both the emitter as well as collector.
 $V_{be} < 0.7$

1.3 CONFIGURATION OF BIPOLAR JUNCTION TRANSISTORS

Since a Bipolar Junction Transistor is a three-terminal device, there are three ways to connect it within an electric circuit while one terminal is the same for both output and input. Every method of connection responds differently to the input signals within a circuit.

- Common Emitter Configuration – has both voltage and current gain
- The common Collector Configuration – has no voltage gain but has a current gain
- The common base configuration – has no current gain but has a voltage gain

Characteristics	Common Base	Common Emitter	Common Collector
Power Gain	low	Very high	medium
Current gain	low	medium	high
Voltage gain	High	Medium	low
Phase angle	0	180	0
Output impedance	Very high	high	low
Input Impedance	Low	medium	high

1.4 APPLICATIONS OF BJT

We know that a bipolar junction transistor is used as a switch, as an amplifier, as a filter, and even as an oscillator. Below is the list of other applications of bipolar junction transistor.

- BJT is used as a detector or also known as a demodulator.
- BJT finds application in clipping circuits so that the waves can be shaped.
- Logic circuits and switching circuits use BJT.

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